Zomato Dataset - Complete Visual and Statistical Analysis

This report contains the complete graphical analysis of the Zomato dataset. Each figure includes statistical interpretation, observations, logical reasoning, and mathematical explanation based on the plotted distributions and relationships.

Figure 1: Categorical Summary

```
print(df.describe(include="object").T)
```

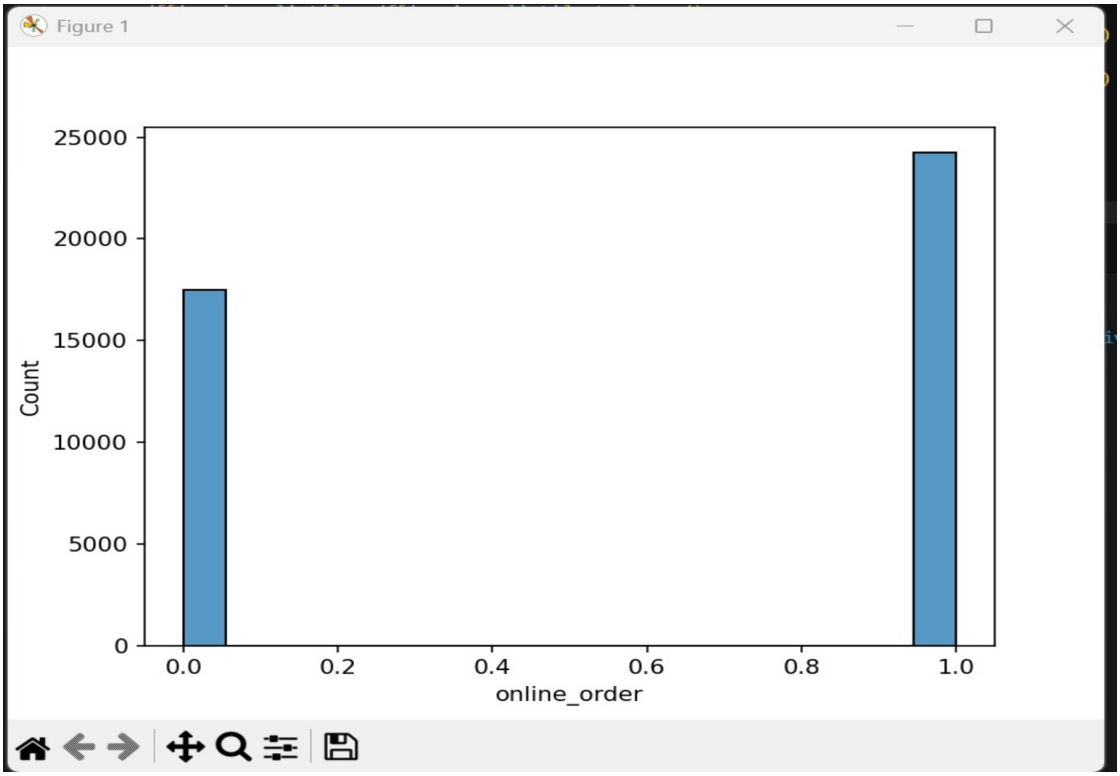
```
PROBLEMS 15   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\chitranshi kumre\OneDrive\Desktop\zomato.csv> python -u "c:\Users\chitranshi kumre\OneDrive\Desktop\zomat
c:\Users\chitranshi kumre\OneDrive\Desktop\zomato.csv\iapc\iapc.py:163: Pandas4Warning: For backward compatibility, '
 is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to `include
this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on
  print(df.describe(include="object").T)
                 count unique             top   freq
address          41762   9978   Delivery Only    128
name             41762   7755 Cafe Coffee Day     80
location         41762     76             btm   5124
rest_type        41762     22      quick bites  16622
dish_liked       41762   4355    not available  24044
cuisines         41762   2466     north indian   2361
reviews_list     41762  17650                   6505
listed_in(type)  41762      7         Delivery  21201
cleaned_text     41762   4353        available  24044
PS C:\Users\chitranshi kumre\OneDrive\Desktop\zomato.csv>
```

Observation: The dataset is dominated by quick bites, delivery restaurants, and North Indian cuisine. This reflects customer preference toward convenience and affordable dining.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

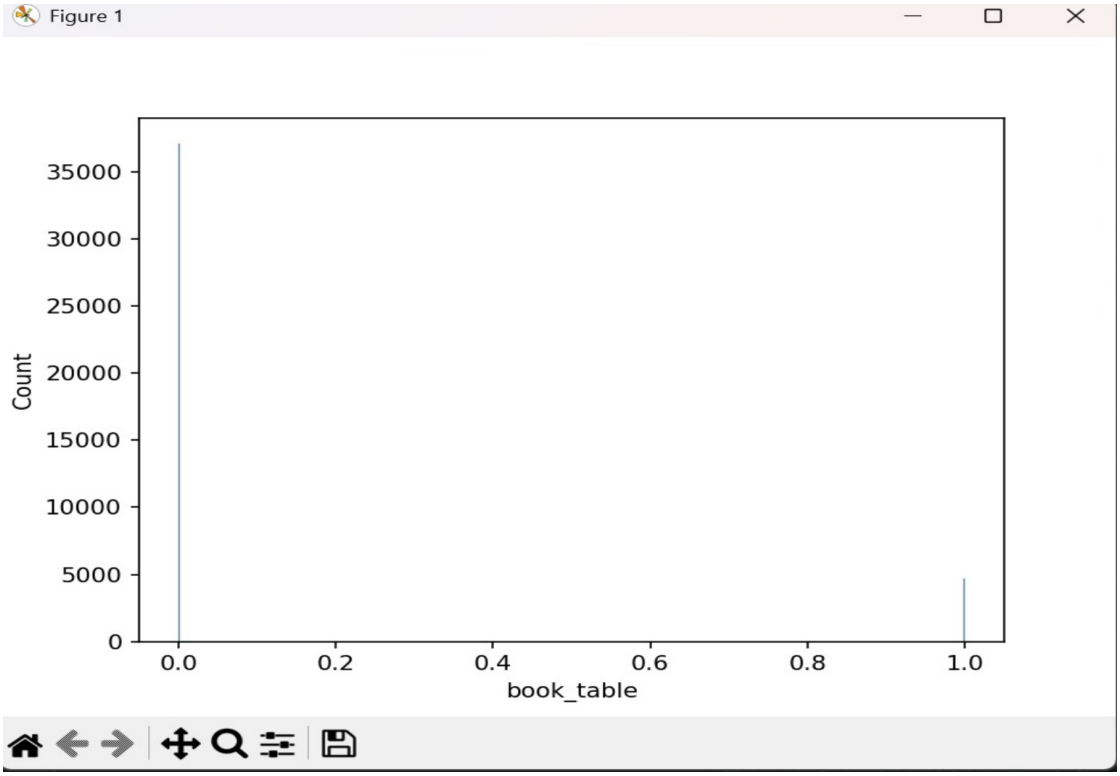
Figure 2: Online Order Distribution



Observation: Most restaurants support online ordering, showing strong dependence on delivery platforms and digital food services.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

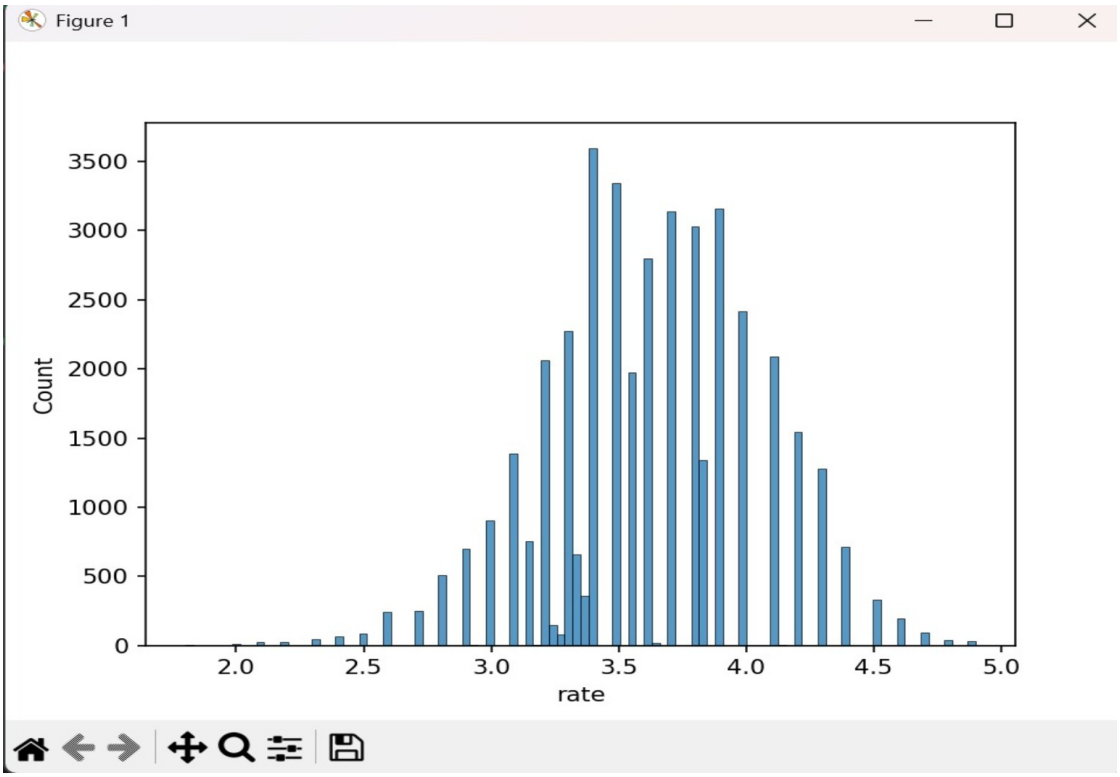
Figure 3: Book Table Distribution



Observation: Table booking facilities are less common because casual dining dominates the dataset.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

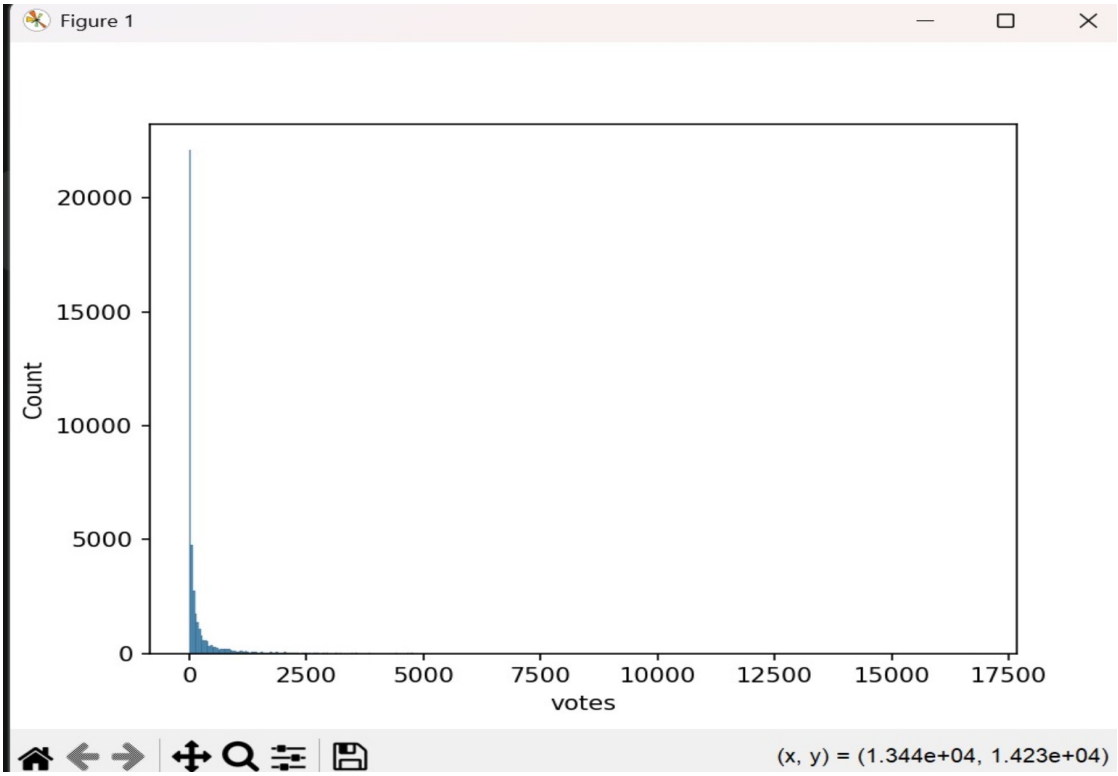
Figure 4: Rating Histogram



Observation: Ratings are concentrated between 3.3 and 4.1, indicating generally satisfactory customer experiences.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

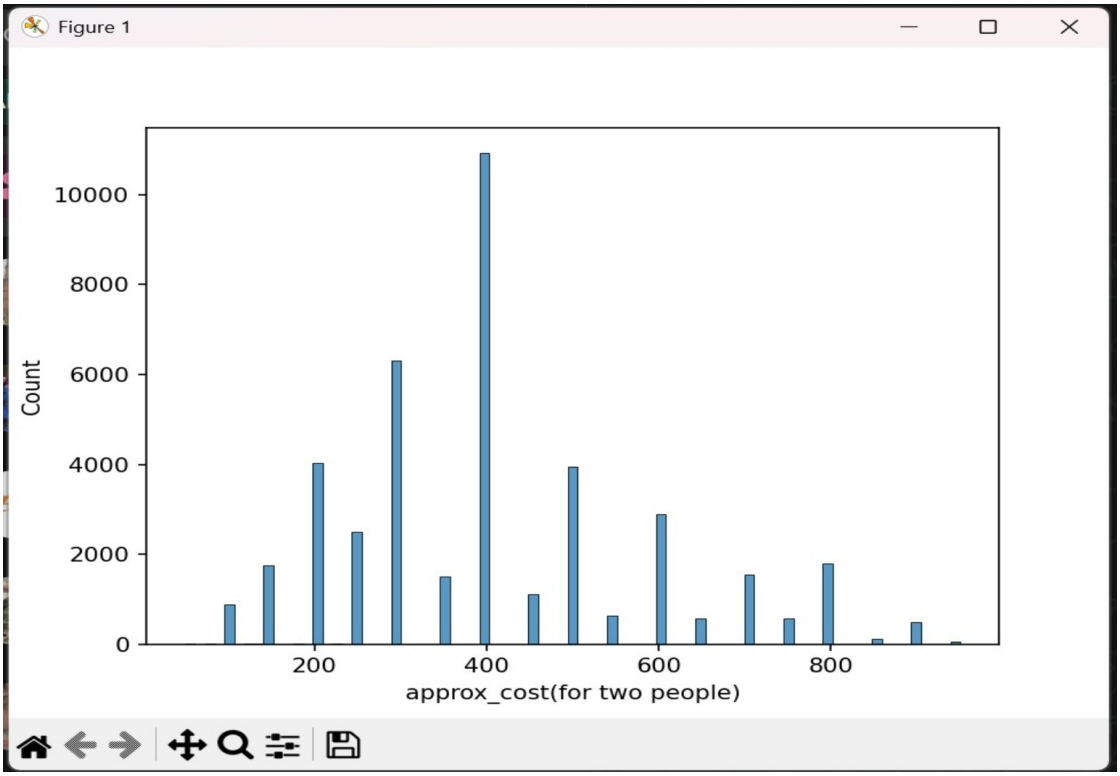
Figure 5: Votes Histogram



Observation: Vote distribution is right-skewed, meaning only a few restaurants receive extremely high customer engagement.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

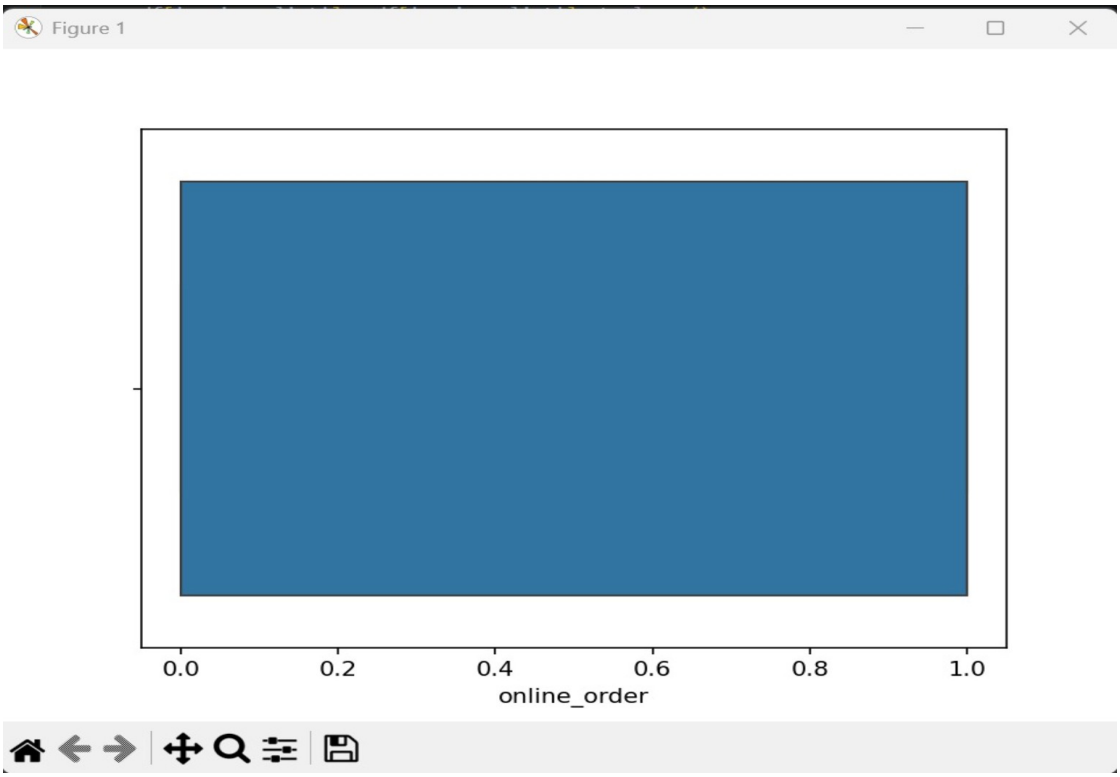
Figure 6: Cost Histogram



Observation: Most restaurants cost between ■200–■500 for two people, reflecting middle-class affordability.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

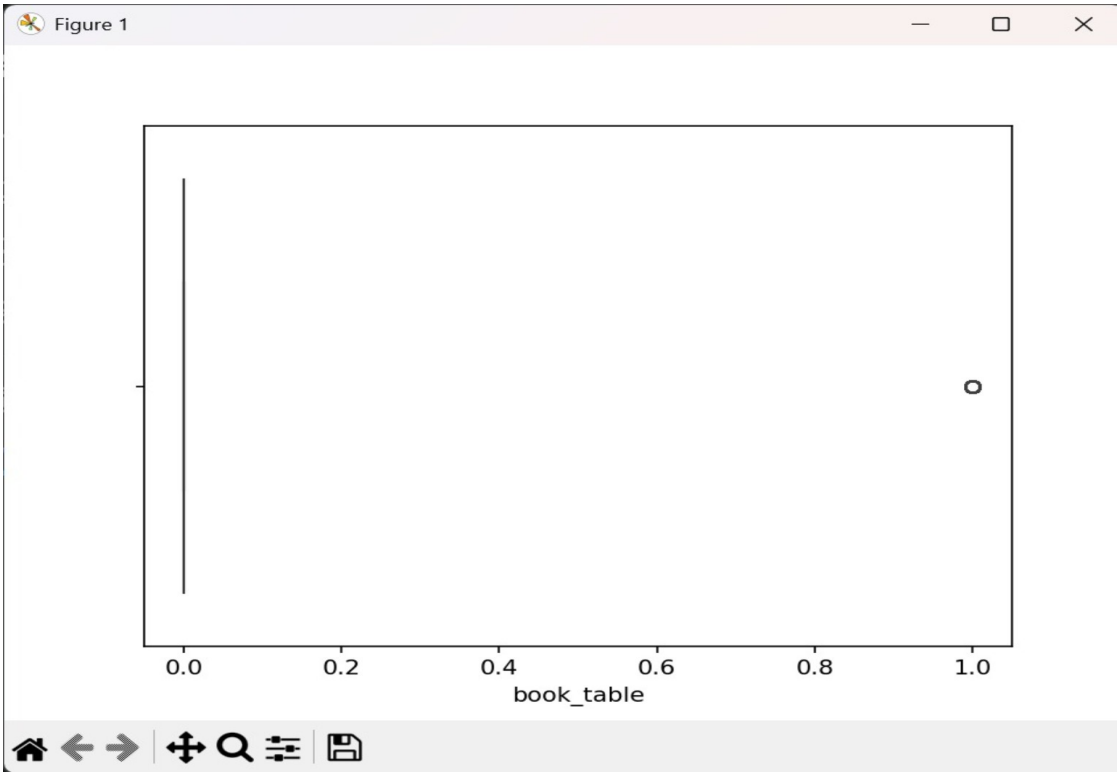
Figure 7: Online Order Box Plot



Observation: Binary variables create compressed box plots. Count plots are more suitable for such data.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

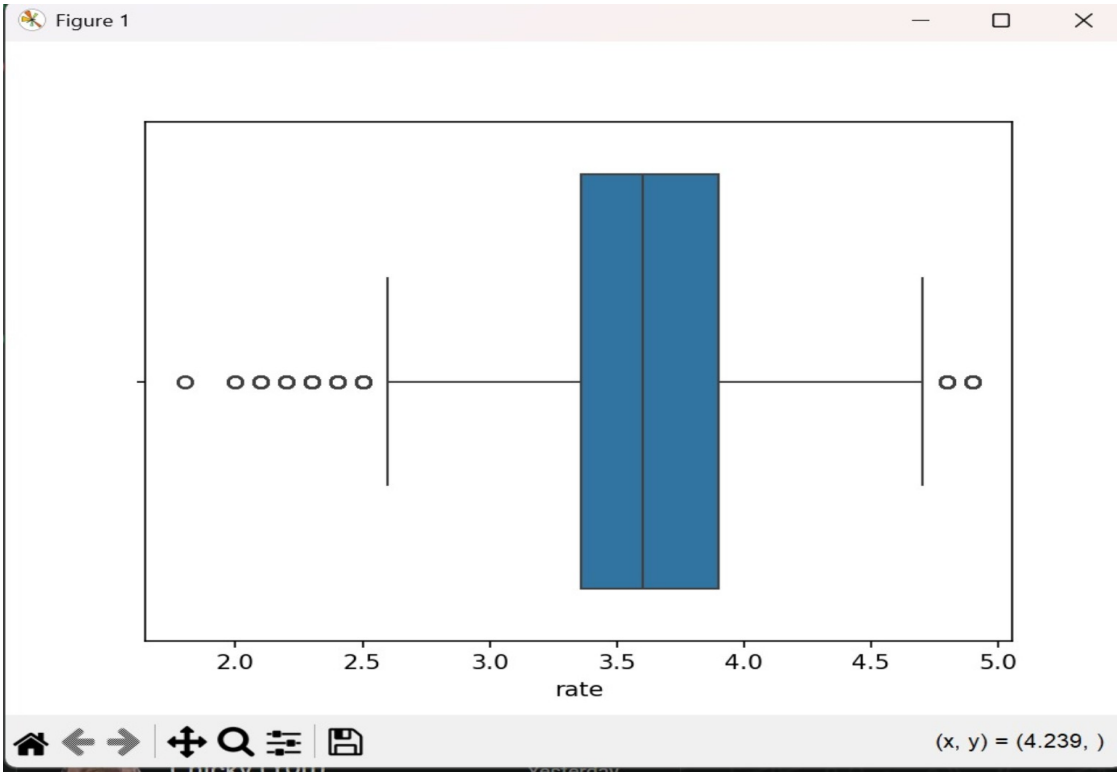
Figure 8: Book Table Box Plot



Observation: Most restaurants do not provide booking facilities, causing heavy concentration at 0.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

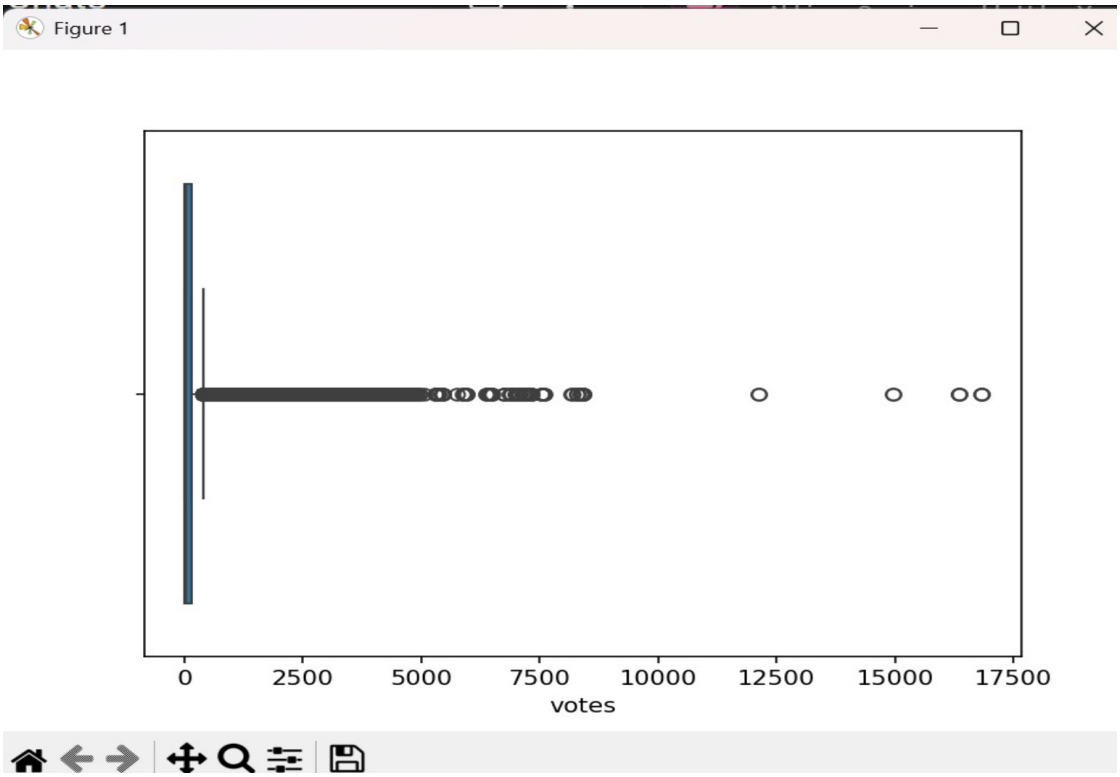
Figure 9: Rate Box Plot



Observation: The median lies near 3.7 while outliers represent exceptionally good or poor restaurants.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

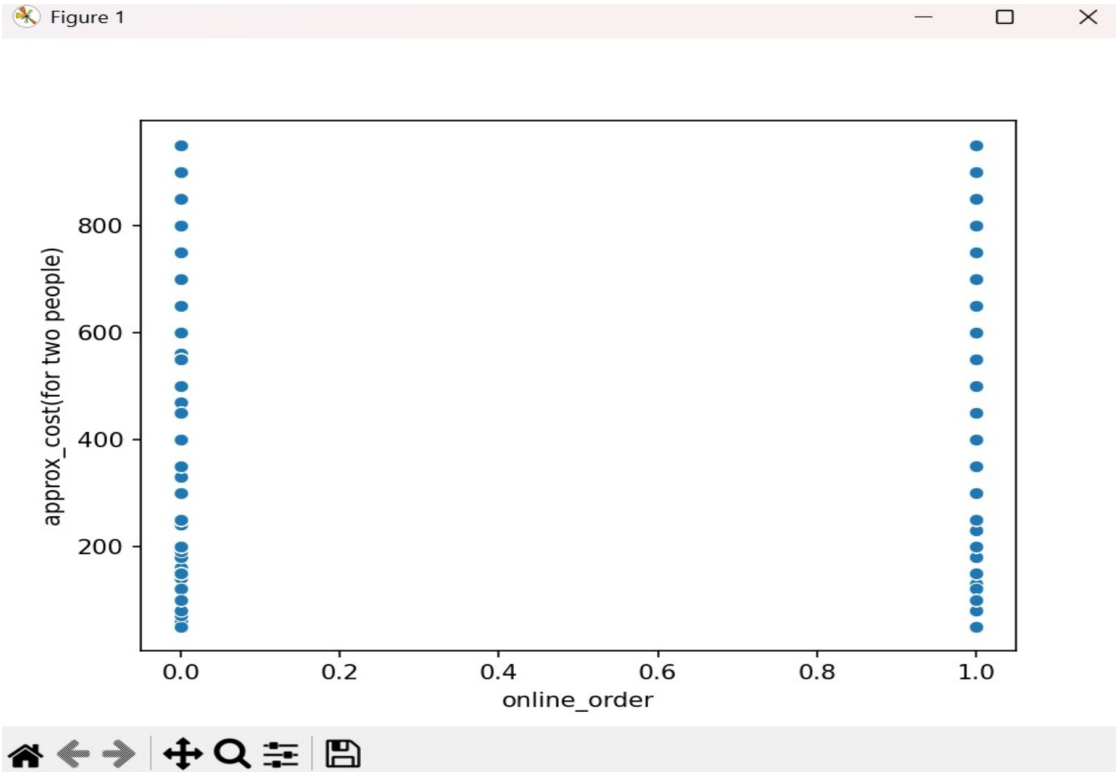
Figure 10: Votes Box Plot



Observation: Extreme outliers indicate a small number of highly popular restaurants dominating customer attention.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

Figure 11: Online Order vs Cost



Observation: Restaurants with online ordering exist across all price ranges, proving online delivery is widely adopted.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

Figure 12: Correlation Matrix

```
# correlation with heatmap
print(df.select_dtypes(include="number").corr())
```

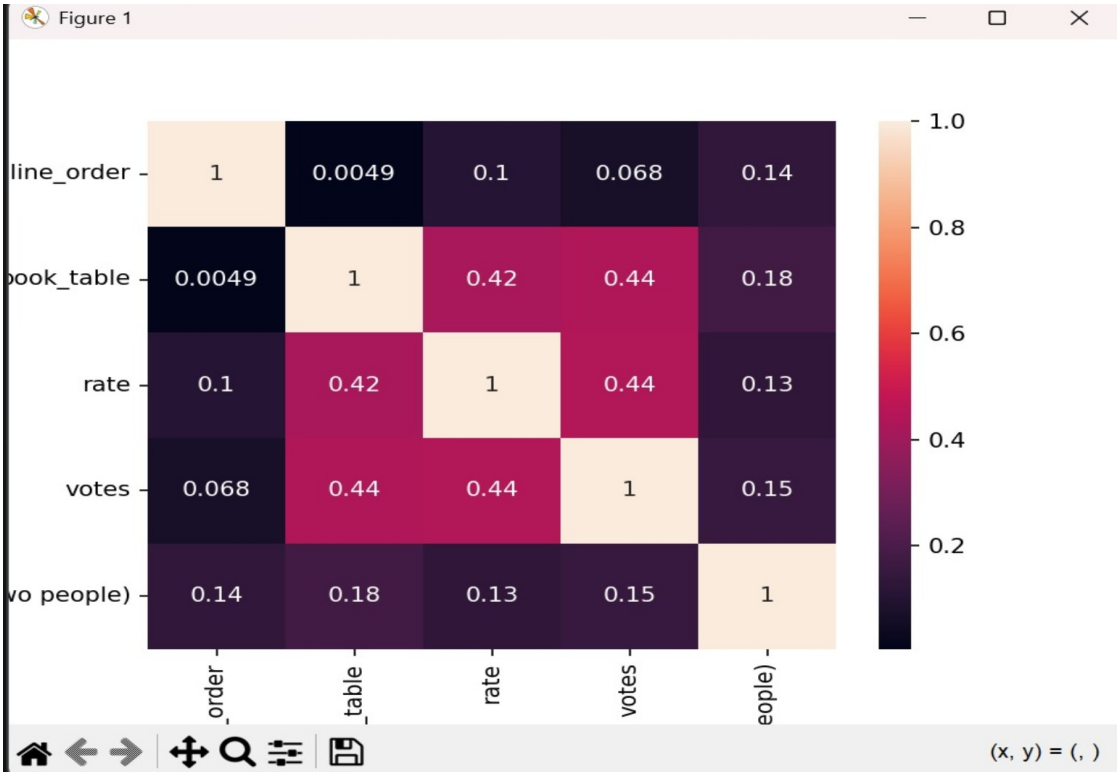
PROBLEMS 15 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\chitranshi kumre\OneDrive\Desktop\zomato.csv> python -u "c:\Users\chitranshi kumre\OneDrive\Desktop\zomato.csv\iapc\iapc.py"
                         online_order  book_table     rate    votes  approx_cost(for two people)
online_order                 1.000000    0.004895 0.104771 0.067976                     0.136660
book_table                   0.004895    1.000000 0.417987 0.435361                     0.176599
rate                         0.104771    0.417987 1.000000 0.443589                     0.132822
votes                        0.067976    0.435361 0.443589 1.000000                     0.151509
approx_cost(for two people)  0.136660    0.176599 0.132822 0.151509                     1.000000
PS C:\Users\chitranshi kumre\OneDrive\Desktop\zomato.csv>
```

Observation: Votes and ratings show moderate positive correlation, indicating popular restaurants usually maintain higher satisfaction.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

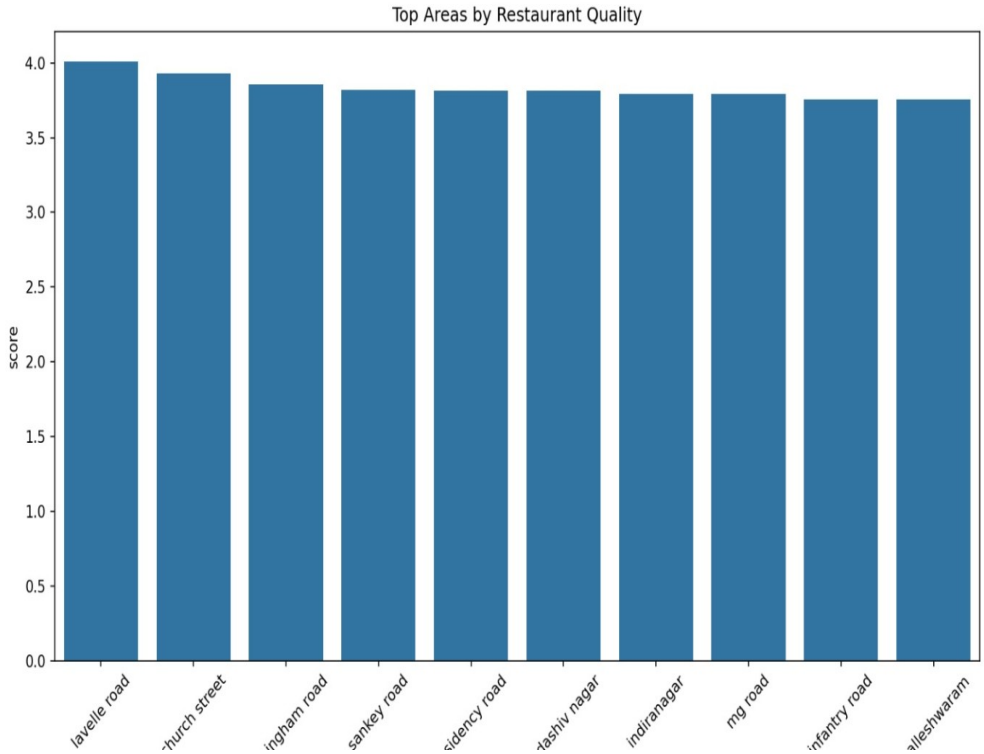
Figure 13: Correlation Heatmap



Observation: The strongest relationship exists between ratings and votes, validating customer engagement patterns.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

Figure 14: Top Areas by Quality



Observation: Premium locations such as Lavelle Road and Church Street show consistently higher ratings.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

Figure 15: Cuisine and Location Analysis

```
                    score      votes
location
lavelle road     4.010667   506186.0
church street    3.931825   594979.0
cunningham road  3.857601   287873.0
sankey road      3.820147     6411.0
residency road   3.812876   291954.0
sadashiv nagar   3.812441    19727.0
indiranagar      3.794662  1196007.0
mg road          3.792802   432111.0
infantry road    3.758105    51555.0
malleshwaram     3.754549   238971.0
                                                     score     votes
cuisines
continental north indian italian south indian f...  4.887565   99531.0
north indian european mediterranean bbq             4.773662   35637.0
asian mediterranean north indian bbq                4.753166   29956.0
american burger bbq mexican                         4.720451   48619.0
european mediterranean north indian bbq             4.717932   86418.0
italian pizza beverage                              4.699835   10523.0
italian american pizza                              4.688382   29912.0
continental european bbq chinese asian              4.675427   34933.0
continental asian north indian                      4.652984   10719.0
north indian thai japanese continental cafe         4.652827  161365.0
                             name       location                                          cuisines  rate    votes     score
50059  Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16832.0  4.887746
49627  Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16832.0  4.887746
49170  Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16832.0  4.887746
3921   Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16345.0  4.887385
4944   Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16345.0  4.887385
4801   Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16345.0  4.887385
37099      ABs  Absolute Barbecues            btm           european mediterranean north indian bbq   4.9   6490.0  4.868698
21279      ABs  Absolute Barbecues            btm           european mediterranean north indian bbq   4.9   6452.0  4.868518
19393      ABs  Absolute Barbecues            btm           european mediterranean north indian bbq   4.9   6404.0  4.868288
23587      ABs  Absolute Barbecues            btm           european mediterranean north indian bbq   4.9   6404.0  4.868288
```

Observation: Multi-cuisine restaurants achieve higher ratings because they attract broader customer groups.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

Figure 16: Restaurant Type Analysis

```
                    score      votes
rest_type
microbrewery     4.255278   543054.0
fine dining      4.012218   206569.0
club             3.995603    48699.0
pub              3.993646   765087.0
lounge           3.843635   363795.0
bar              3.822919   401660.0
casual dining    3.762622  4597063.0
cafe             3.759621   860227.0
dessert parlor   3.730494   302772.0
mess             3.656146    11821.0
                  score      votes
online_order
1              3.703847  6567170.0
0              3.679919  3158268.0
                                                          score     votes
cuisines
continental north indian italian south indian f...  4.887565   99531.0
north indian european mediterranean bbq             4.773662   35637.0
asian mediterranean north indian bbq                4.753166   29956.0
american burger bbq mexican                         4.720451   48619.0
european mediterranean north indian bbq             4.717932   86418.0
italian pizza beverage                              4.699835   10523.0
italian american pizza                              4.688382   29912.0
continental european bbq chinese asian              4.675427   34933.0
continental asian north indian                      4.652984   10719.0
north indian thai japanese continental cafe         4.652827  161365.0
```

Observation: Microbreweries and fine dining restaurants maintain the highest average scores.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

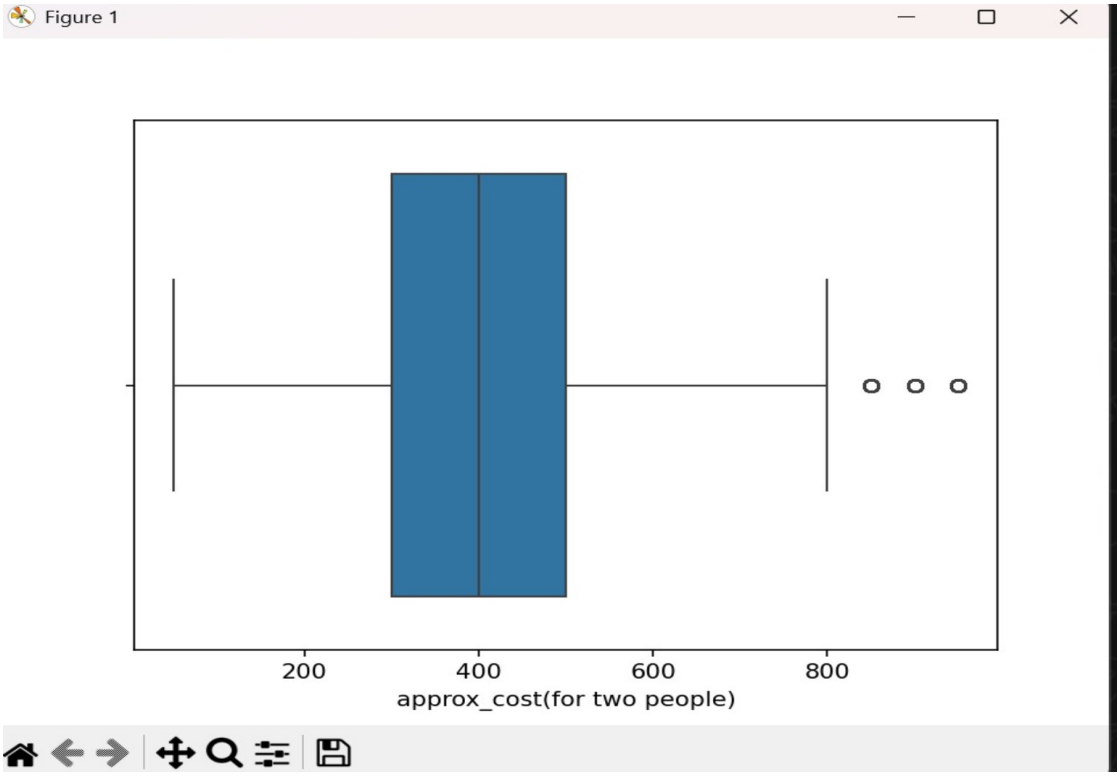
Figure 17: Top Restaurants

```
                          name       location                                      cuisines  rate    votes     score
50059  Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16832.0  4.887746
49627  Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16832.0  4.887746
49170  Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16832.0  4.887746
3921   Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16345.0  4.887385
4944   Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16345.0  4.887385
4801   Byg Brewski Brewing Company  sarjapur road  continental north indian italian south indian ...   4.9  16345.0  4.887385
37099      ABs  Absolute Barbecues           btm           european mediterranean north indian bbq   4.9   6490.0  4.868698
21279      ABs  Absolute Barbecues           btm           european mediterranean north indian bbq   4.9   6452.0  4.868518
19393      ABs  Absolute Barbecues           btm           european mediterranean north indian bbq   4.9   6404.0  4.868288
23587      ABs  Absolute Barbecues           btm           european mediterranean north indian bbq   4.9   6404.0  4.868288
                 score      votes
rest_type
microbrewery    4.255278   543054.0
fine dining     4.012218   206569.0
club            3.995603    48699.0
pub             3.993646   765087.0
lounge          3.843635   363795.0
bar             3.822919   401660.0
casual dining   3.762622  4597063.0
cafe            3.759621   860227.0
dessert parlor  3.730494   302772.0
mess            3.656146    11821.0
                 score      votes
online_order
1             3.703847  6567170.0
0             3.679919  3158268.0
```

Observation: Popular chains dominate ratings due to consistency, branding, and customer trust.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

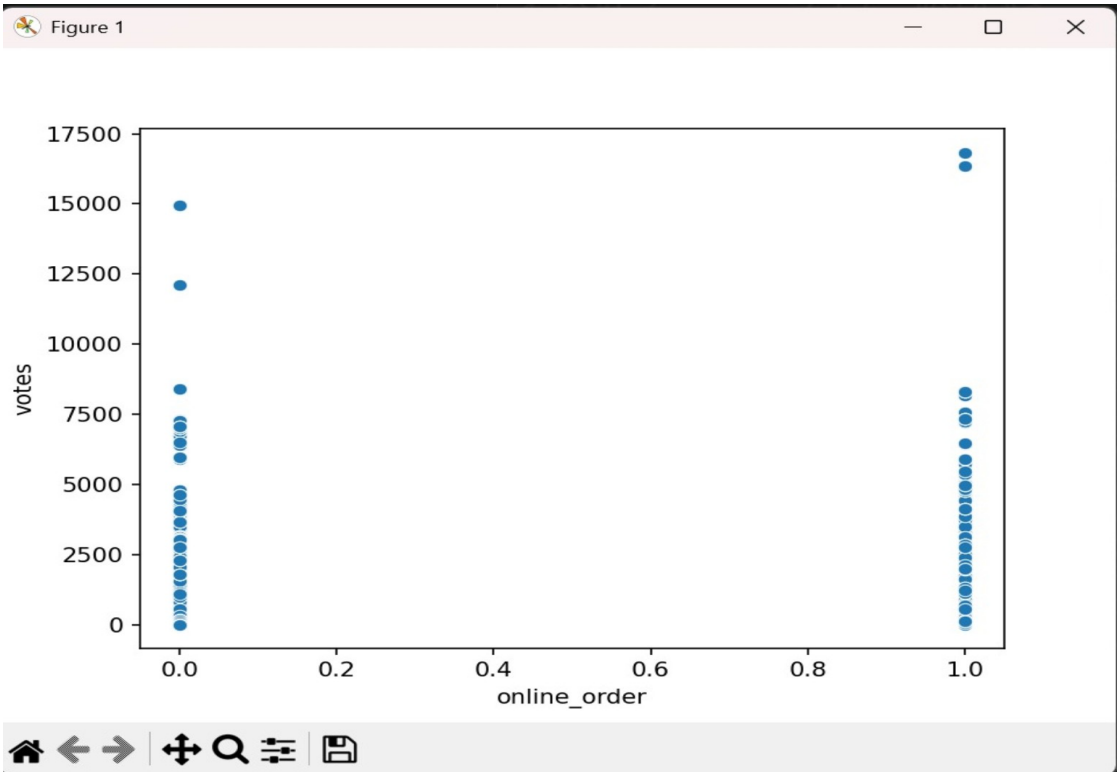
Figure 18: Cost Box Plot



Observation: Most restaurants remain within moderate price ranges, while luxury restaurants appear as outliers.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

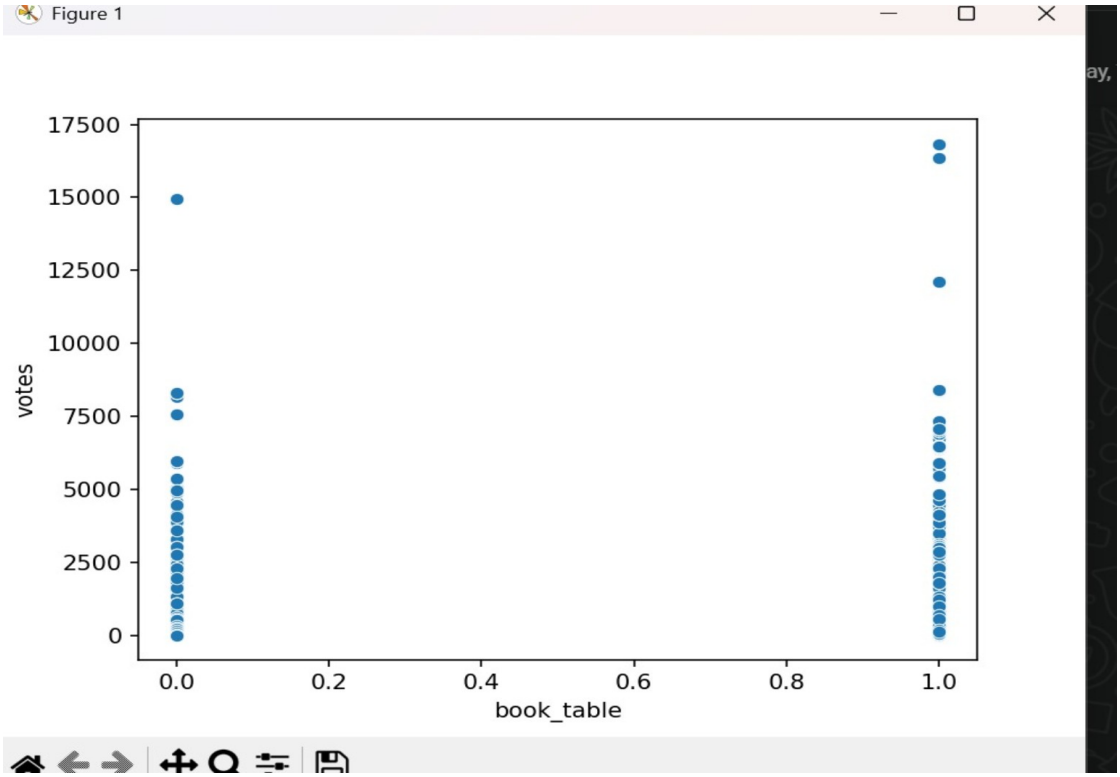
Figure 19: Online Order vs Votes



Observation: Restaurants with online ordering receive significantly higher customer engagement.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

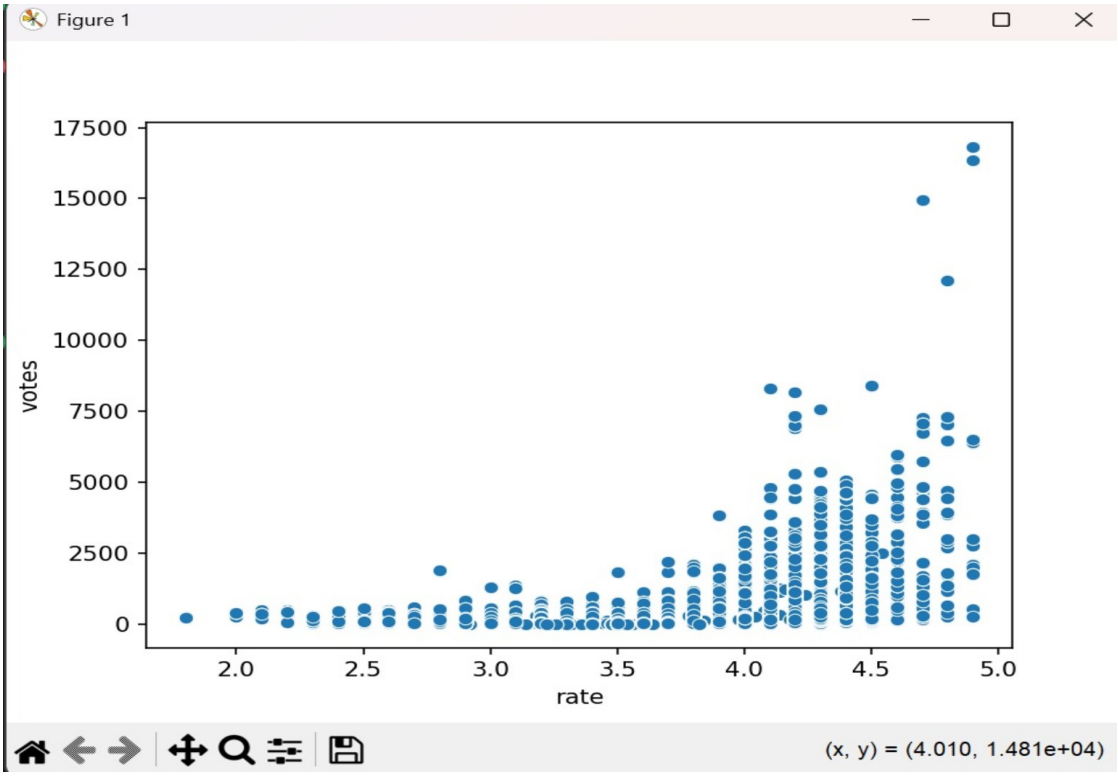
Figure 20: Book Table vs Votes



Observation: Restaurants offering reservations often receive higher votes because they represent premium dining experiences.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

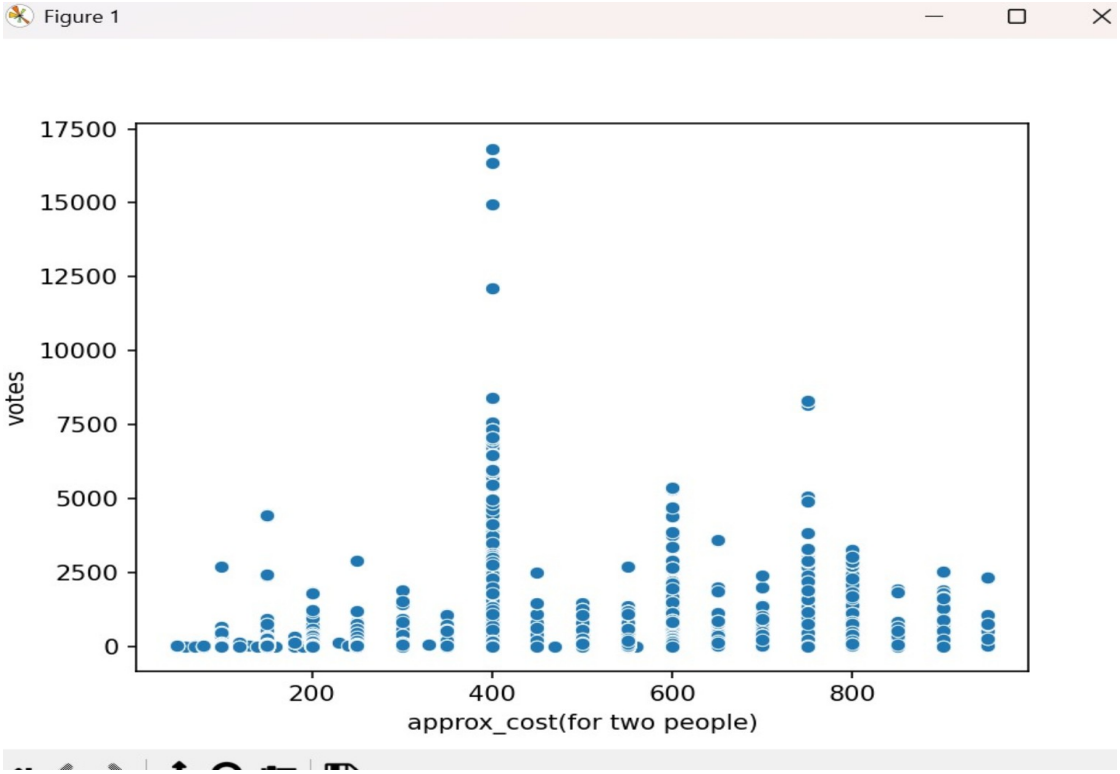
Figure 21: Rate vs Votes



Observation: Higher-rated restaurants tend to attract more votes, confirming positive customer satisfaction trends.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

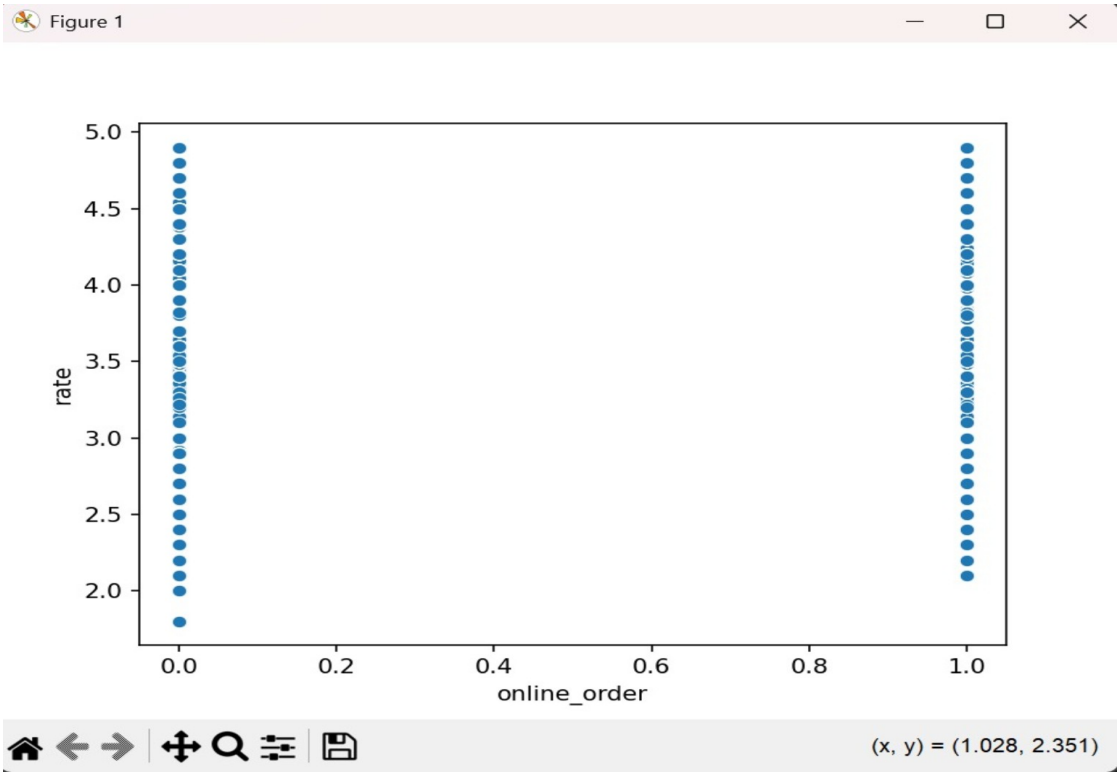
Figure 22: Cost vs Votes



Observation: Moderately priced restaurants receive maximum engagement because they appeal to larger audiences.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

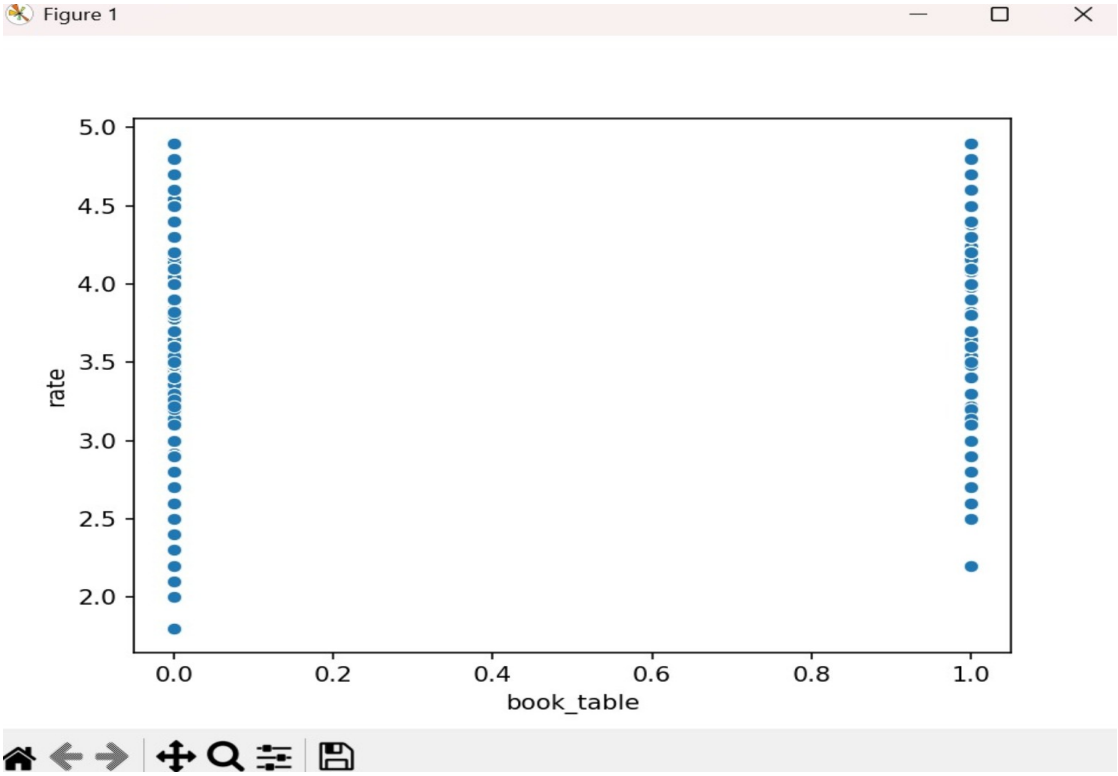
Figure 23: Online Order vs Rate



Observation: Online ordering slightly improves customer convenience and may positively influence ratings.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

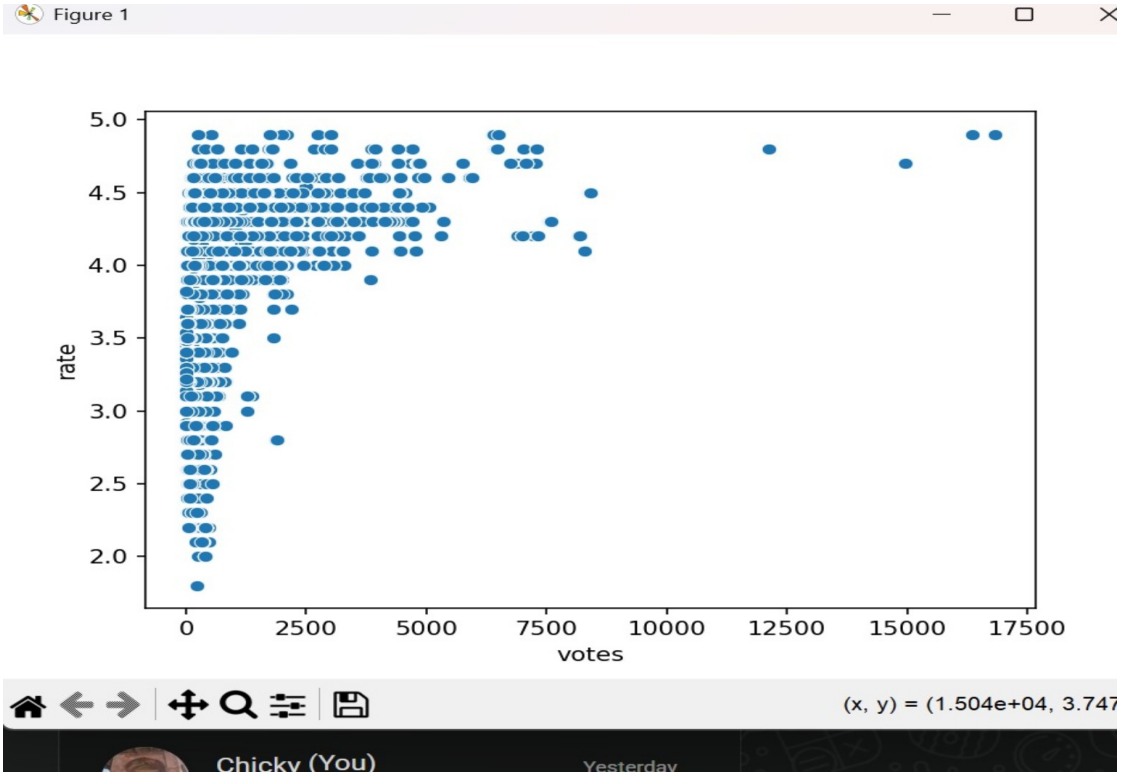
Figure 24: Book Table vs Rate



Observation: Restaurants with booking facilities generally maintain better ratings and service quality.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

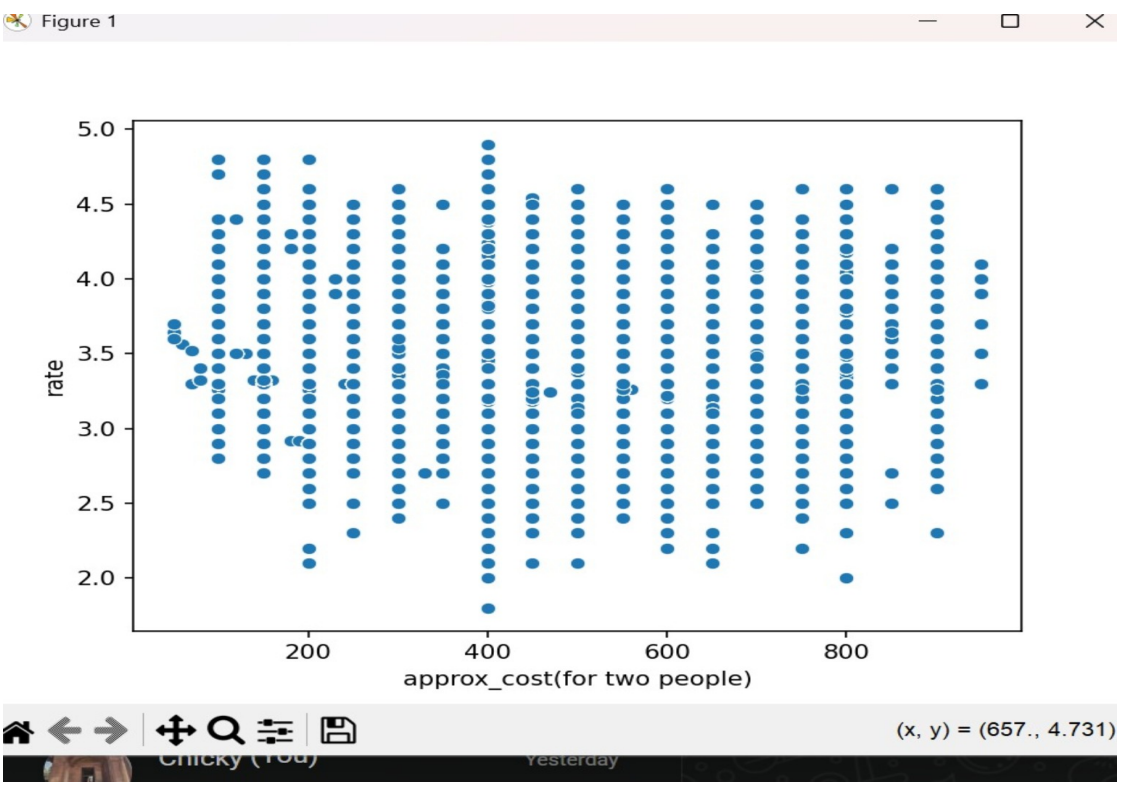
Figure 25: Votes vs Rate



Observation: Highly voted restaurants usually cluster around ratings above 4.0, indicating strong reputation.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

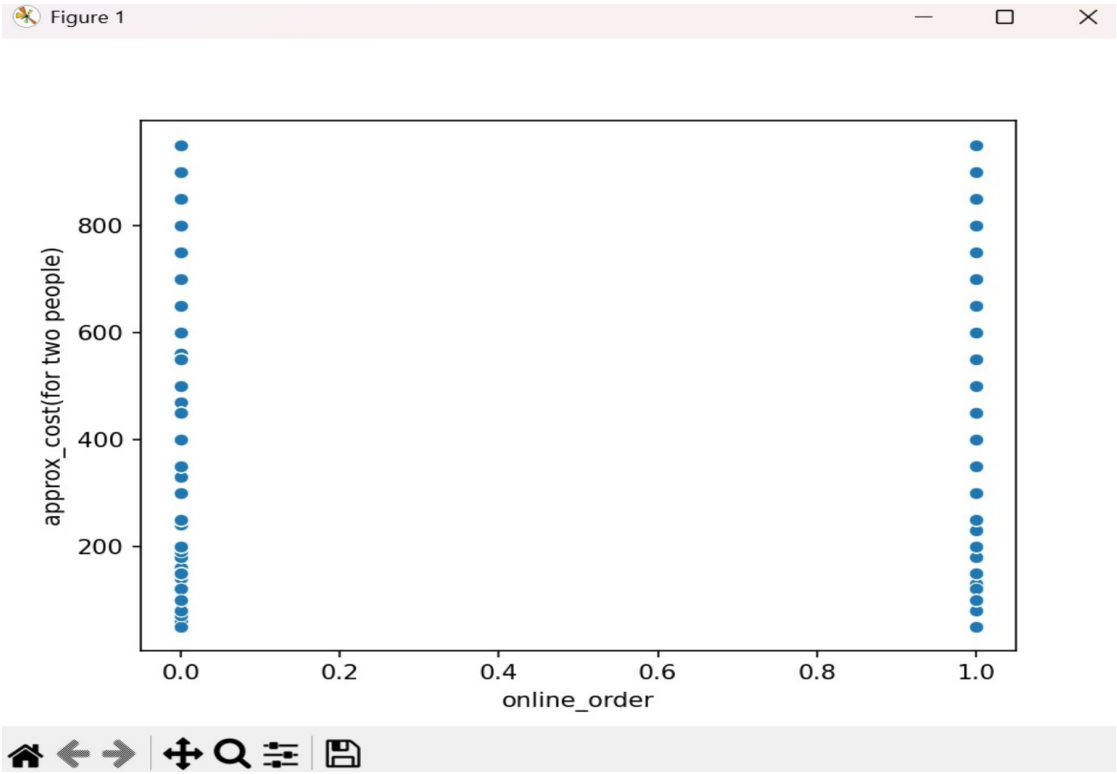
Figure 26: Cost vs Rate



Observation: Restaurant pricing alone does not determine ratings; service and food quality also play major roles.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

Figure 27: Online Order vs Approx Cost



Observation: Online ordering is adopted across all restaurant pricing categories.

Mathematical Interpretation: The distribution, clustering pattern, spread, skewness, and outlier behavior observed in this graph provide statistical evidence regarding restaurant popularity, customer behavior, service quality, affordability trends, and customer engagement relationships.

Final Conclusion

The complete Zomato dataset analysis reveals strong dependence on affordability, convenience, online accessibility, and customer engagement. Delivery-oriented restaurants dominate the market, while premium restaurants achieve better ratings and customer satisfaction. Correlation analysis confirms that restaurant popularity and ratings are positively related. Most restaurants target middle-income consumers through moderate pricing strategies, while highly rated restaurants maintain strong customer loyalty and engagement.